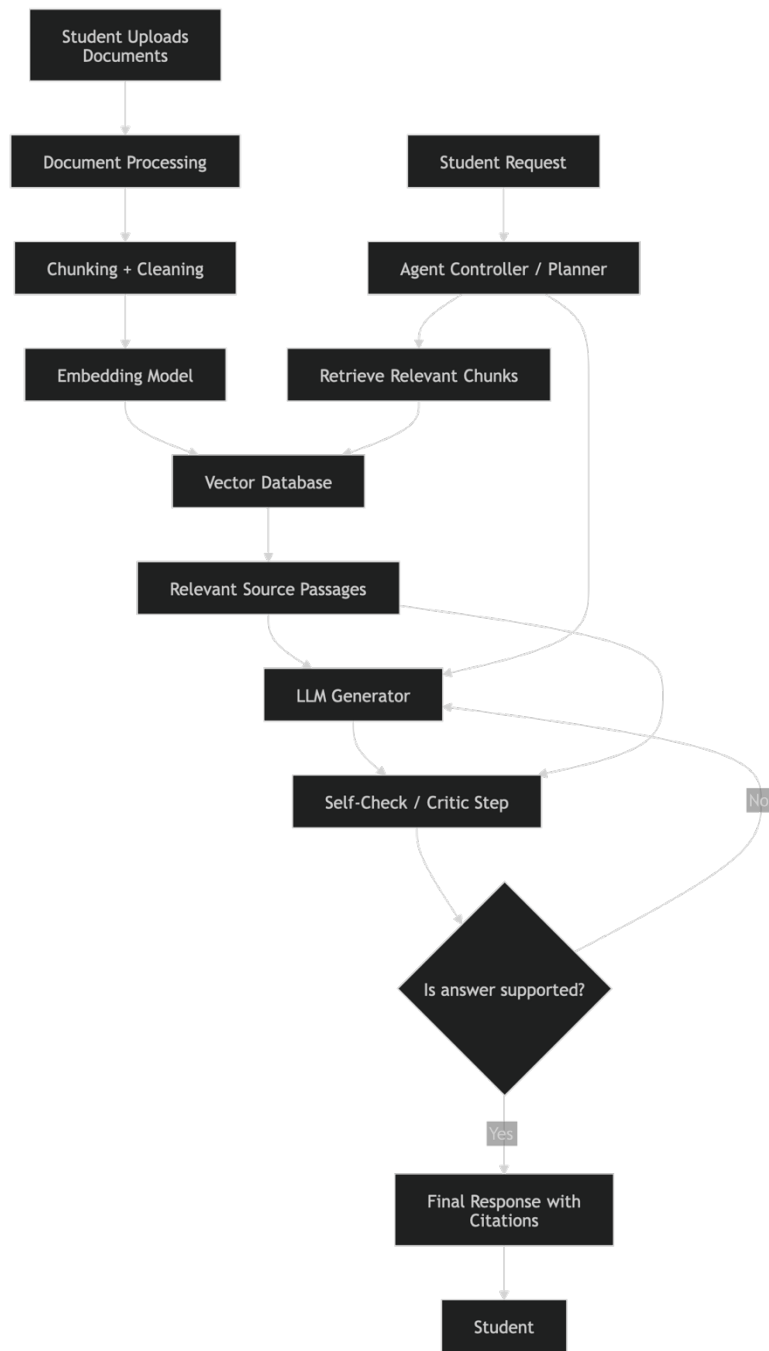


StudyBud

An agentic learning system that plans study workflows, generates materials, and verifies its outputs against source documents using iterative self-correction.

System Architecture:



Simple Explanation

The student uploads course materials. The system breaks those documents into chunks, turns them into embeddings, and stores them in a vector database.

When the student asks for a quiz, essay, summary, or explanation, the **agent controller** decides what to do. It retrieves relevant document sections, sends them to the LLM, generates the study material, then checks whether the answer is actually supported by the documents before giving it to the student.

The key agentic loop is:

Retrieve → Generate → Check → Revise → Respond

Features

1. Document Upload

Allow students to upload PDFs, notes, or text files.

2. RAG Search

The system should retrieve relevant chunks from the uploaded documents.

3. Study Content Generator

Support 3 basic actions:

- Summarize this topic
- Generate practice questions
- Explain this concept

4. Source-Based Answers

The answer should include citations like:

Based on page 4, the paper explains that...

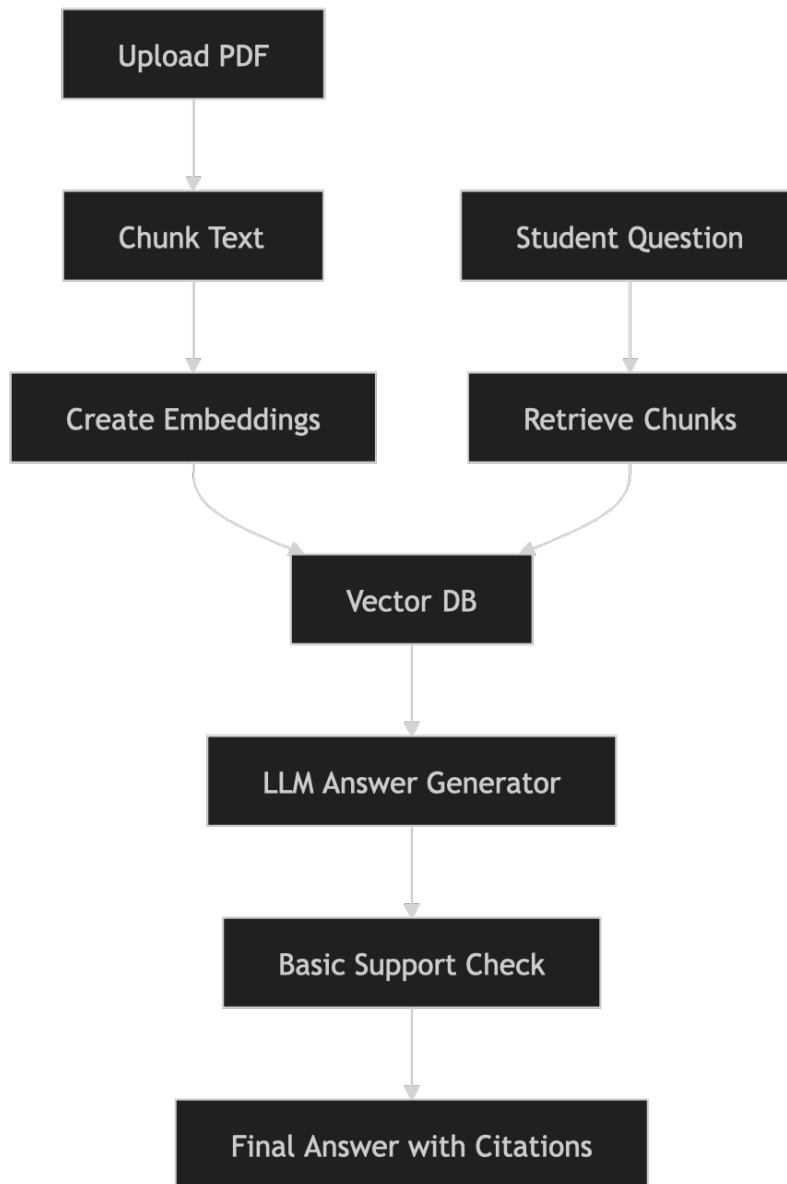
5. Basic Self-Check

Before responding, the agent asks:

Is this answer supported by the retrieved text?

If not, it revises the answer.

Architecture:



Goal

Should answer:

Can a student upload course material and get useful, document-grounded study help?